

Directional Relational Manifolds: Stratified Variable-Rank Geometry, Rank-Transition Holonomy, Geometric Hysteresis, and Energy-Controlled Transitions

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Abstract

We introduce Directional Relational Manifolds (DRM), a stratified variable-rank geometric framework for systems whose effective degrees of freedom may emerge, collapse, or reorganize with state. The construction starts from a smooth finite-rank vector bundle $E \rightarrow M$, a smooth symmetric positive-semidefinite bundle metric g , and an anchor $\rho : E \rightarrow TM$. The effective relational fiber and effective dimension are

$$\overline{E}_p = E_p / \text{Ker } g_p, \quad d_{\text{DRM}}(p) = \text{rank } g_p.$$

The base is decomposed into constant-rank strata. On each stratum, the effective quotient fibers form a smooth metric vector bundle on which connections, curvature, parallel transport, energy, and admissible dynamics are defined. Rank-changing trajectories are modeled as hybrid geometric paths: invertible metric transport acts inside strata, while possibly non-invertible transition maps act at rank-changing events.

The global contribution is a theory of *rank-transition holonomy*. For a closed hybrid path, the ordered composition of intrastatum parallel transports and interstratum transition maps defines an endomorphism of the initial effective fiber. Unlike ordinary holonomy, this operator need not be invertible. We prove that if a closed loop based at an n -dimensional effective fiber contains a transition of rank $r < n$, then the total hybrid holonomy has rank at most r and therefore cannot be the identity. Thus a trajectory may return to the same base point and the same nominal dimension without returning to the same relational state. This produces *geometric hysteresis*: path-dependent memory generated by dimensional collapse and reactivation. The effect may occur even when every regular stratum is flat and every intrastatum transport is trivial.

We further establish an energy criterion for rank transitions. A transition is dissipative exactly when its metric operator norm is at most one, or, equivalently in matrix form, $J^\top G_+ J \preceq G_-$. Compositions of dissipative transitions and metric parallel transports are globally non-expansive and generate a contraction semigroup rather than a holonomy group. Rank defect, norm defect, energy loss, spectral persistence, and transition commutators are introduced as quantitative hysteresis invariants. Explicit examples exhibit irreversible rank-2 $\rightarrow$ 1 $\rightarrow$ 2 cycles, order-dependent collapse maps, and memory in flat strata. Under constant-rank, nondegeneracy, anchor-isomorphism, and compatibility hypotheses, DRM reduces to Riemannian geometry; with an injective anchor it reduces to sub-Riemannian geometry; and with the Fisher information metric it reduces to Fisher–Rao information geometry.

Keywords: directional relational manifolds; variable-rank geometry; degenerate metric; stratified geometry; rank-transition holonomy; geometric hysteresis; non-invertible transport; hybrid dynamics; emergent dimensionality; information geometry.

1 Introduction

Classical differential geometry normally fixes local dimension before dynamics is introduced. A smooth n -manifold has tangent spaces of constant dimension n , and a Riemannian metric is nondegenerate at every point. These assumptions are appropriate for many systems, but they do not directly encode situations in which the number of effective, distinguishable, controllable, or dynamically available directions changes with state.

Examples include constrained mechanical systems, synchronization and mode collapse, loss and recovery of controllable directions, rank-deficient statistical models, changing identifiability, adaptive representations, and hybrid systems whose internal transition structure reorganizes during evolution.

Directional Relational Manifolds are proposed as a controlled framework for such systems. The ambient capacity is represented by a fixed-rank vector bundle, while the effective geometry is determined by the rank of a positive-semidefinite relational metric. Directions in the metric kernel are relationally indistinguishable at the given state. Quotienting by that kernel produces the effective fiber.

The theory is deliberately stratified. Smooth differential geometry is used only on constant-rank strata. Rank changes are not forced into a fictitious globally smooth connection; instead, they are represented by explicit transition maps. This produces a hybrid geometry consisting of smooth intrastatum evolution and discrete interstratum reorganization.

The decisive global observation is that a closed base-space loop need not produce an invertible map of the initial effective fiber. If information is lost during an intermediate dimensional collapse, deterministic later expansion cannot reconstruct that information from the surviving state alone. The resulting closed-loop transport is therefore naturally semigroup-valued, not group-valued.

1.1 Main contributions

No novelty is claimed for positive-semidefinite metrics, rank stratifications, quotient bundles, anchors, connections, rank inequalities, or hybrid reset maps considered separately. The contribution is their integration into the chain

$$\begin{aligned} \text{variable metric rank} &\longrightarrow \text{effective relational dimension} \\ &\longrightarrow \text{stratumwise quotient geometry} \\ &\longrightarrow \text{hybrid rank-transition transport} \\ &\longrightarrow \text{non-invertible holonomy and hysteresis.} \end{aligned}$$

The principal results are:

- (i) a precise stratified DRM structure based on effective quotient fibers;
- (ii) an ordered hybrid transport law for paths visiting multiple rank strata;
- (iii) a transition-holonomy semigroup for closed hybrid loops;
- (iv) a dimensional holonomy theorem forcing rank-deficient loop transport whenever a genuine rank-reducing transition occurs;
- (v) a proof that geometric hysteresis can exist with zero intrastatum curvature;

- (vi) energy and matrix criteria for dissipative, isometric, and expansive rank transitions;
- (vii) quantitative memory invariants based on rank, norm, energy, spectrum, and path order;
- (viii) reduction theorems connecting DRM to established geometries.

1.2 Organization

Section 2 positions DRM relative to existing frameworks. Sections 3–4 define the ambient bundle, relational metric, quotient fibers, and rank stratification. Sections 5–6 introduce anchored motion, connections, curvature, and regular transport. Sections 7–8 define hybrid paths and rank-transition holonomy. Sections 9–11 develop geometric hysteresis, energy control, and order dependence. Section 13 introduces relational action. Sections 14 and 15 give examples and reduction theorems.

2 Relation to Existing Mathematical Frameworks

2.1 Degenerate metrics

A smooth metric may become degenerate, preventing use of the ordinary inverse metric and obstructing standard Levi–Civita constructions. Singular semi-Riemannian geometry develops curvature-like objects under additional regularity hypotheses [10]. DRM takes a different route: on each constant-rank stratum it quotients the ambient fiber by the metric kernel, thereby obtaining a positive-definite effective metric. It does not claim to replace intrinsic theories of degenerate metrics; it supplies a quotient-and-transition framework adapted to changing effective rank.

2.2 Stratified spaces, singular distributions, and holonomy groupoids

Stratified geometry studies spaces decomposed into smooth pieces of different dimensions. Singular distributions and singular foliations permit pointwise variation in rank and possess sophisticated orbit and holonomy theories [11, 2]. DRM instead stratifies a smooth ambient base by the rank of a semidefinite bundle metric and assigns an effective quotient bundle to each stratum. Its interstratum maps are explicit model data and may be non-invertible. Consequently, the global transport object is generally a category or semigroup of linear maps, rather than a groupoid of locally invertible arrows.

2.3 Anchored bundles and Lie algebroids

An anchored vector bundle consists of a vector bundle and a bundle map into the tangent bundle. A Lie algebroid adds a compatible bracket on sections [8]. DRM uses anchors to convert internal relational directions into observable state velocities. A Lie-algebroid structure may be imposed on a regular stratum when intrinsic torsion and curvature are needed, but it is not part of the minimal global definition.

2.4 Sub-Riemannian geometry

Sub-Riemannian geometry equips a distribution of admissible directions with a positive-definite metric. On a constant-rank stratum with injective effective anchor, a DRM induces a sub-Riemannian structure. DRM additionally permits the effective distribution rank to change through metric degeneration and incorporates explicit reset maps between rank strata.

2.5 Information geometry

Information geometry equips regular statistical models with the Fisher metric [1, 3, 9]. Rank-deficient Fisher matrices occur in non-identifiable or singular models. DRM interprets the metric rank as an effective relational dimension and treats passages through singular rank loci using stratification and transition maps. Fisher–Rao geometry is recovered in the regular constant-rank case.

2.6 Hybrid dynamical systems

Hybrid systems combine continuous flows with discrete jumps or reset maps [4]. DRM uses this architecture geometrically: connections and flows act inside rank strata, while transition maps act on effective fibers at rank-changing events. The maps are assessed not only dynamically but also metrically, through energy inequalities.

2.7 Classical holonomy

For a connection on a fixed-rank vector bundle, parallel transport around a closed loop gives an invertible endomorphism, and the holonomy at a base point forms a subgroup of $\mathrm{GL}(E_p)$ [5, 7]. In DRM, regular transport remains invertible inside each stratum, but interstratum maps may lower rank. The ordered closed-loop transport can therefore lie in $\mathrm{End}(\overline{E}_p) \setminus \mathrm{GL}(\overline{E}_p)$. The qualifier “rank-transition” distinguishes this object from ordinary holonomy.

3 Ambient Relational Structure

Definition 3.1 (Ambient state manifold). *Let M be a smooth, Hausdorff, second-countable manifold.*

Definition 3.2 (Ambient relational bundle). *Let $\pi : E \rightarrow M$ be a smooth real vector bundle of finite constant rank N .*

The rank N is the maximum ambient relational capacity. It is not the effective local dimension.

Definition 3.3 (Relational metric). *A relational metric is a smooth section*

$$g \in \Gamma(\mathrm{Sym}^2 E^*)$$

such that every bilinear form $g_p : E_p \times E_p \rightarrow \mathbb{R}$ is symmetric and positive semidefinite.

Definition 3.4 (Relational kernel). *At $p \in M$, define*

$$K_p := \mathrm{Ker} g_p = \{v \in E_p : g_p(v, w) = 0 \text{ for every } w \in E_p\}.$$

Definition 3.5 (Effective relational fiber). *The effective fiber at p is*

$$\overline{E}_p := E_p / K_p.$$

Proposition 3.1 (Induced effective metric). *The formula*

$$\overline{g}_p([v], [w]) := g_p(v, w)$$

defines a positive-definite inner product on $\overline{E}_p$.

Proof. If $v' = v + k_1$ and $w' = w + k_2$ with $k_1, k_2 \in K_p$, every term containing k_1 or k_2 vanishes, so the value is independent of representatives. If $\bar{g}_p([v], [v]) = 0$, the Cauchy–Schwarz inequality for positive semidefinite forms gives $g_p(v, w) = 0$ for every w , hence $v \in K_p$ and $[v] = 0$. $\square$

Definition 3.6 (Effective relational dimension). *The DRM dimension at p is*

$$d_{\text{DRM}}(p) := \text{rank } g_p = \dim \bar{E}_p.$$

Remark 3.1. *Dimension is not the cardinality of a directional generating set. Rank is the invariant measuring the number of metrically distinguishable directions.*

4 Rank Stratification

For $r \in \{0, \dots, N\}$, define

$$M_r := \{p \in M : \text{rank } g_p = r\}.$$

Assumption 4.1 (Regular rank stratification). *Each connected component of every nonempty M_r is a smooth embedded submanifold of M , and the collection of components is locally finite.*

A connected component $S \subseteq M_r$ is called a rank- r stratum.

Proposition 4.1 (Kernel subbundle). *On a rank- r stratum S , the union*

$$K|_S := \bigcup_{p \in S} K_p$$

is a smooth vector subbundle of $E|_S$ of rank $N - r$.

Proof. The metric induces a smooth bundle morphism

$$g^\flat : E|_S \rightarrow E^*|_S, \quad g_p^\flat(v) = g_p(v, \cdot).$$

Its rank is constantly r on S . The constant-rank theorem for vector-bundle morphisms [6] implies that $\text{Ker } g^\flat = K|_S$ is a smooth subbundle. $\square$

Corollary 4.1 (Effective quotient bundle). *The quotient*

$$\bar{E}^{(S)} := E|_S / K|_S$$

is a smooth rank- r vector bundle over S , equipped with the smooth positive-definite metric $\bar{g}^{(S)}$.

The quotient bundles generally do not assemble into a single vector bundle over all of M , because their ranks differ.

5 Anchor and Admissible Motion

Definition 5.1 (Anchor). *An anchor is a smooth vector-bundle morphism $\rho : E \rightarrow TM$ covering the identity on M .*

Assumption 5.1 (Kernel compatibility and stratum tangency). *For every $p \in M$,*

$$K_p \subseteq \text{Ker } \rho_p.$$

Moreover, for every rank stratum S and every $p \in S$,

$$\rho_p(E_p) \subseteq T_p S \subseteq T_p M.$$

Proposition 5.1 (Effective stratum anchor). *Kernel compatibility defines a unique map*

$$\bar{\rho}_p : \bar{E}_p \rightarrow T_p M, \quad \bar{\rho}_p([v]) = \rho_p(v).$$

On every stratum S , the tangency assumption makes these maps a smooth bundle morphism

$$\bar{\rho}^{(S)} : \bar{E}^{(S)} \rightarrow TS.$$

Consequently, $\bar{\rho}(U)[f]$ is intrinsically defined for $U \in \Gamma(\bar{E}^{(S)})$ and $f \in C^\infty(S)$.

Definition 5.2 (Admissible curve). *Let S be a stratum. A C^1 curve $\gamma : I \rightarrow S$ is admissible if there exists a smooth field $V(t) \in \bar{E}_{\gamma(t)}^{(S)}$ such that*

$$\dot{\gamma}(t) = \bar{\rho}_{\gamma(t)} V(t).$$

The effective anchor may be neither injective nor surjective. Internal directions can be unobservable, and tangent directions can be inaccessible.

6 Stratumwise Connections, Curvature, and Regular Transport

Let S be a fixed rank stratum.

Assumption 6.1 (Anchored bracket). *The bundle $\bar{E}^{(S)}$ is equipped with a bracket*

$$[\cdot, \cdot]_S : \Gamma(\bar{E}^{(S)}) \times \Gamma(\bar{E}^{(S)}) \rightarrow \Gamma(\bar{E}^{(S)})$$

satisfying

$$[U, fV]_S = f[U, V]_S + \bar{\rho}(U)[f]V, \quad \bar{\rho}([U, V]_S) = [\bar{\rho}(U), \bar{\rho}(V)].$$

If the Jacobi identity also holds, the stratum carries a Lie-algebroid structure.

Definition 6.1 (Relational connection). *A relational connection on $\bar{E}^{(S)}$ is a map*

$$\nabla^{(S)} : \Gamma(\bar{E}^{(S)}) \times \Gamma(\bar{E}^{(S)}) \rightarrow \Gamma(\bar{E}^{(S)})$$

that is $C^\infty(S)$ -linear in the first argument and satisfies

$$\nabla_U^{(S)}(fV) = \bar{\rho}(U)[f]V + f\nabla_U^{(S)}V.$$

Definition 6.2 (Metric compatibility). *The connection is metric-compatible when*

$$\bar{\rho}(U)[\bar{g}(V, W)] = \bar{g}(\nabla_U^{(S)}V, W) + \bar{g}(V, \nabla_U^{(S)}W).$$

Definition 6.3 (Torsion and curvature). *Define*

$$T^{(S)}(U, V) = \nabla_U^{(S)} V - \nabla_V^{(S)} U - [U, V]_S$$

and

$$R^{(S)}(U, V)W = \nabla_U^{(S)} \nabla_V^{(S)} W - \nabla_V^{(S)} \nabla_U^{(S)} W - \nabla_{[U, V]_S}^{(S)} W.$$

Proposition 6.1 (Existence of metric-compatible connections). *Every effective metric bundle $(\bar{E}^{(S)}, \bar{g}^{(S)})$ admits a metric-compatible connection.*

Proof. Choose any connection ∇^0 and define

$$Q(U; V, W) = \bar{\rho}(U)[\bar{g}(V, W)] - \bar{g}(\nabla_U^0 V, W) - \bar{g}(V, \nabla_U^0 W).$$

Because $\bar{g}$ is nondegenerate, there is a unique endomorphism-valued one-form A satisfying

$$2\bar{g}(A_U V, W) = Q(U; V, W).$$

Then $\nabla_U V := \nabla_U^0 V + A_U V$ is metric-compatible. □

A metric-compatible connection defines parallel transport

$$P_{s \rightarrow t}^\gamma : \bar{E}_{\gamma(s)}^{(S)} \rightarrow \bar{E}_{\gamma(t)}^{(S)}$$

along admissible curves in S . It is an isometry and therefore invertible. For a closed loop ℓ in S based at p ,

$$\text{Hol}_\ell^{(S)} := P_{0 \rightarrow 1}^\ell \in \text{GL}(\bar{E}_p)$$

is the ordinary stratumwise holonomy.

7 Hybrid Paths and Ordered Transport

Definition 7.1 (Rank-transition event). *Let $p_- \in S_-$ and $p_+ \in S_+$ lie in strata of ranks r and s , respectively. A rank-transition event consists of the ordered base reset $p_- \rightsquigarrow p_+$ together with a linear map*

$$J_{r \rightarrow s} : \bar{E}_{p_-} \rightarrow \bar{E}_{p_+}.$$

The points p_- and p_+ need not coincide, and the map need not be invertible. Thus an event records both the reset of the base state and the reset of the effective relational state.

Definition 7.2 (Hybrid DRM path). *A hybrid DRM path is a finite sequence*

$$\gamma = (\gamma_1, J_1, \gamma_2, J_2, \dots, J_{m-1}, \gamma_m),$$

where each $\gamma_i : [a_i, b_i] \rightarrow S_i$ is a smooth admissible path contained in one rank stratum and each J_i is a transition event

$$J_i : \bar{E}_{\gamma_i(b_i)} \longrightarrow \bar{E}_{\gamma_{i+1}(a_{i+1})}.$$

No continuity of the base path is imposed at a transition event. A continuous crossing of a rank boundary can instead be treated only after specifying one-sided trace fibers and limiting parallel transports; those additional analytic data are not assumed in the present finite-event framework.

Let P_i denote parallel transport along γ_i .

Definition 7.3 (Ordered hybrid transport). *The hybrid transport operator is*

$$\mathcal{P}_\gamma = P_m J_{m-1} P_{m-1} \cdots J_2 P_2 J_1 P_1.$$

The order is essential. The maps may be noncommuting and may act between spaces of different dimensions.

Proposition 7.1 (Concatenation law). *If γ ends where η begins, then*

$$\mathcal{P}_{\eta*\gamma} = \mathcal{P}_\eta \mathcal{P}_\gamma.$$

Proof. A vector is first transported along γ and then along η . This is exactly composition of the ordered operators. $\square$

Remark 7.1 (Path category). *If rank strata are represented as vertices and permitted transitions as directed edges, hybrid paths form a path category. Sending each vertex to its effective fiber and each path to its ordered transport defines a functor into the category of finite-dimensional vector spaces and linear maps. The target is not generally a groupoid because transition maps need not be invertible.*

8 Rank-Transition Holonomy

Definition 8.1 (Closed hybrid loop). *A hybrid path γ is closed and based at p if its initial and final base points coincide with p , and the initial and final effective fibers are identified with $\overline{E}_p$.*

Definition 8.2 (Rank-transition holonomy). *For a closed hybrid loop γ based at p , define*

$$H_\gamma := \mathcal{P}_\gamma \in \text{End}(\overline{E}_p).$$

Unlike ordinary holonomy, H_γ need not belong to $\text{GL}(\overline{E}_p)$.

Definition 8.3 (Transition-holonomy semigroup). *The transition-holonomy semigroup at p is*

$$\text{THol}_p(\mathfrak{D}) := \{H_\gamma : \gamma \text{ is a closed hybrid DRM loop based at } p\} \subseteq \text{End}(\overline{E}_p).$$

Proposition 8.1. *$\text{THol}_p(\mathfrak{D})$ contains the identity and is closed under composition. It need not contain inverses.*

Proof. The constant loop gives the identity. Concatenation of closed loops gives composition by the concatenation law. If a loop holonomy has deficient rank, it is not invertible and no inverse belongs to $\text{End}(\overline{E}_p)$ as an inverse map. $\square$

Theorem 8.1 (Dimensional holonomy theorem). *Let γ be a closed hybrid DRM loop based at p , with $\dim \overline{E}_p = n$. Suppose one transition map J_k appearing in the ordered composition has rank $r < n$. Then*

$$\text{rank } H_\gamma \leq r < n.$$

Consequently,

$$H_\gamma \neq \text{Id}_{\overline{E}_p},$$

and H_γ is non-invertible.

Proof. Write the ordered composition as

$$H_\gamma = AJ_k B,$$

where B contains every map applied before J_k and A contains every map applied after it. The rank inequality for compositions gives

$$\text{rank}(AJ_k B) \leq \text{rank}(J_k) = r.$$

The identity on an n -dimensional space has rank n , so H_γ cannot be the identity and cannot be invertible. $\square$

Corollary 8.1 (Irreversible dimensional cycle). *A closed loop containing a genuine rank-reducing transition cannot recover the identity transport by later deterministic linear maps acting only on the surviving effective state.*

Proof. Once the rank of the total map has fallen below n , postcomposition cannot increase it above the rank bound established in Theorem 8.1. $\square$

Remark 8.1 (External memory). *Recovery is possible only after enlarging the state with an auxiliary channel that stores the discarded information. The enlarged evolution may be invertible even though the projected effective evolution is not.*

9 Geometric Hysteresis and Memory Invariants

Definition 9.1 (Geometric hysteresis). *A closed hybrid loop exhibits geometric hysteresis if*

$$H_\gamma \neq \text{Id}.$$

It exhibits irreversible geometric hysteresis if H_γ is non-invertible.

A system exhibiting hysteresis returns to the same base point but not to the same effective relational state.

Definition 9.2 (Surviving and lost subspaces). *For a closed loop γ , define*

$$\mathcal{M}_\gamma := \text{im } H_\gamma, \quad \mathcal{L}_\gamma := \text{Ker } H_\gamma.$$

Definition 9.3 (Rank-memory defect). *If $\dim \overline{E}_p = n$, define*

$$\delta_{\text{rank}}(\gamma) := n - \text{rank } H_\gamma.$$

Definition 9.4 (Norm-memory defect). *Define*

$$\delta_{\text{op}}(\gamma) := \|H_\gamma - \text{Id}\|_{\text{op}}.$$

A normalized Frobenius alternative is

$$h_F(\gamma) := \frac{\|H_\gamma - \text{Id}\|_F}{\sqrt{n}}.$$

Theorem 9.1 (Hysteresis without intrastatum curvature). *Let γ be a closed hybrid DRM loop. Assume:*

- (i) every regular stratum visited by γ is flat along its regular segments;
- (ii) every intrastatum transport is the identity;
- (iii) at least one transition map has rank below the initial fiber dimension n .

Then H_γ is non-invertible and different from the identity.

Proof. The identity transports drop out, leaving an ordered product of transition maps. Theorem 8.1 applies. $\square$

Remark 9.1. *This separates two sources of path dependence:*

$$\text{intrastatum curvature} \quad \text{and} \quad \text{interstratum dimensional memory}.$$

The second can remain nontrivial when the first vanishes.

10 Energy-Controlled Rank Transitions

Let the incoming and outgoing effective fibers be finite-dimensional inner product spaces

$$(V_-, \langle \cdot, \cdot \rangle_-), \quad (V_+, \langle \cdot, \cdot \rangle_+).$$

Definition 10.1 (Transition energy). *For $v \in V_-$, define*

$$\mathcal{E}_-(v) = \frac{1}{2} \|v\|_-^2, \quad \mathcal{E}_+(Jv) = \frac{1}{2} \|Jv\|_+^2.$$

Definition 10.2 (Transition classes). *A transition $J : V_- \rightarrow V_+$ is:*

- (i) dissipative if $\|Jv\|_+ \leq \|v\|_-$ for every v ;
- (ii) isometric if equality holds for every v ;
- (iii) expansive if the dissipative inequality fails for some v .

Define the metric operator norm

$$\|J\|_{-\rightarrow+} := \sup_{v \neq 0} \frac{\|Jv\|_+}{\|v\|_-}.$$

Proposition 10.1 (Transition-energy criterion). *A transition J is dissipative if and only if*

$$\|J\|_{-\rightarrow+} \leq 1.$$

It is isometric if and only if

$$J^* \bar{g}_+ = \bar{g}_-.$$

Proof. The first assertion is the definition of the induced operator norm. Equality of norms for every vector is equivalent, by polarization, to preservation of the full inner product, which is $J^* \bar{g}_+ = \bar{g}_-$. $\square$

Corollary 10.1 (Rank-loss obstruction to isometry). *If $\dim V_+ < \dim V_-$, no map $J : V_- \rightarrow V_+$ is isometric on all of V_- .*

Corollary 10.2 (Canonical dissipative collapse). *Let $W \subseteq V_-$ and let $\Pi_W : V_- \rightarrow W$ be the orthogonal projection. If $V_+ = W$ with the restricted metric, then Π_W is dissipative and*

$$\|v\|_-^2 = \|\Pi_W v\|_+^2 + \|v - \Pi_W v\|_-^2.$$

Proposition 10.2 (Compositional stability). *A composition of dissipative transitions is dissipative.*

Proof. The induced operator norm is submultiplicative. $\square$

Theorem 10.1 (Global non-expansiveness). *Consider a hybrid path composed of metric-compatible parallel transports P_i and dissipative transitions J_i . Then*

$$\|\mathcal{P}_\gamma v\|_{\text{final}} \leq \|v\|_{\text{initial}}$$

for every initial vector v .

Proof. Every metric parallel transport has norm one and every dissipative transition has norm at most one. Apply submultiplicativity to the ordered product. $\square$

Corollary 10.3 (Contraction semigroup). *Dissipative closed-loop holonomies form a contraction semigroup in $\text{End}(\overline{E}_p)$.*

Definition 10.3 (Energy hysteresis). *For a closed loop based at p , define*

$$\Delta_\gamma(v) := \frac{1}{2}\|v\|^2 - \frac{1}{2}\|H_\gamma v\|^2.$$

For dissipative loops, $\Delta_\gamma(v) \geq 0$.

In a basis with incoming metric matrix G_- and outgoing metric matrix G_+ , dissipativity is equivalent to

$$J^\top G_+ J \preceq G_-.$$

For a closed loop with metric matrix G , define the energy-defect matrix

$$D_\gamma := G - H_\gamma^\top G H_\gamma.$$

The loop is dissipative exactly when $D_\gamma \succeq 0$.

Remark 10.1 (Dimension activation). *When $\dim V_+ > \dim V_-$, a transition may be isometric on its image, but the new complementary directions are not determined by the incoming vector. Forcing, noise, initialization data, a learned rule, or an auxiliary memory channel is needed to populate them.*

11 Path-Order Dependence and Noncommutative Memory

Definition 11.1 (Transition commutator). *For endomorphisms A, B of a common effective fiber, define*

$$[A, B]_{\text{tr}} := AB - BA.$$

Definition 11.2 (Order-memory defect). *Define*

$$\delta_{\text{ord}}(A, B) := \|AB - BA\|,$$

and the normalized index

$$h_{\text{ord}}(A, B) := \frac{\|AB - BA\|_F}{\max\{1, \|A\|_F \|B\|_F\}}.$$

Proposition 11.1 (Noncommutative dimensional memory). *If $[A, B]_{\text{tr}} \neq 0$, then there exists an initial vector for which the ordered histories AB and BA produce different final states.*

Proof. A nonzero linear map $AB - BA$ acts nontrivially on at least one vector. $\square$

Thus the same local transition operations can encode different memory when traversed in a different order.

12 Spectral and Asymptotic Memory

Repeated traversal of a closed loop evolves the state by

$$v_k = H_\gamma^k v_0.$$

Proposition 12.1 (Strict contraction). *If $\|H_\gamma\|_{\text{op}} < 1$, then $H_\gamma^k v \rightarrow 0$ for every v .*

Proof. $\|H_\gamma^k v\| \leq \|H_\gamma\|_{\text{op}}^k \|v\| \rightarrow 0$. $\square$

Proposition 12.2 (Projection memory). *If $H_\gamma^2 = H_\gamma$, then*

$$H_\gamma^k v = H_\gamma v \quad (k \geq 1).$$

The persistent memory space is $\text{im } H_\gamma$.

More generally, the spectrum and Jordan structure of H_γ determine asymptotic persistence, decay, and possible transient amplification.

13 Relational Energy, Length, and Hybrid Action

Let $\gamma : [a, b] \rightarrow S$ be admissible in one stratum and satisfy $\dot{\gamma} = \bar{\rho}(V)$.

Definition 13.1 (Relational energy and length). *Define*

$$\mathcal{E}[\gamma, V] = \frac{1}{2} \int_a^b \bar{g}(V, V) dt, \quad L[\gamma, V] = \int_a^b \sqrt{\bar{g}(V, V)} dt.$$

Definition 13.2 (Stratumwise relational geodesic). *An admissible pair (γ, V) is a stratumwise relational geodesic if it is stationary for $\mathcal{E}$ under admissible fixed-endpoint variations.*

For a hybrid path with segments (γ_i, V_i) and transitions J_i , define

$$\mathcal{A}_{\text{hyb}} = \sum_i \mathcal{E}[\gamma_i, V_i] + \sum_i C_i(v_i^-, v_i^+),$$

where a natural cost for a dissipative transition is

$$C_i = \frac{1}{2} \|v_i^-\|_-^2 - \frac{1}{2} \|J_i v_i^-\|_+^2 \geq 0.$$

This separates continuous action inside strata from energy lost at discrete rank changes.

14 Examples

14.1 A corrected rank-2 $\rightarrow 1 \rightarrow 2$ cycle

Let the initial and final effective fibers be $\mathbb{R}^2$. Define

$$J_{2 \rightarrow 1}(a, b) = a, \quad J_{1 \rightarrow 2}(a) = (a, 0),$$

and assume all regular transports are identities. The closed-loop holonomy is

$$H = J_{1 \rightarrow 2} J_{2 \rightarrow 1} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

Therefore

$$H(a, b) = (a, 0), \quad \text{rank } H = 1, \quad \delta_{\text{rank}} = 1.$$

The energy loss is

$$\Delta(a, b) = \frac{1}{2}(a^2 + b^2) - \frac{1}{2}a^2 = \frac{1}{2}b^2.$$

The map is idempotent, so one traversal projects the state onto a persistent one-dimensional memory subspace.

14.2 A smooth metric realizing a rank boundary

Let

$$M = \mathbb{R}, \quad E = M \times \mathbb{R}^2, \quad g_x = \begin{pmatrix} 1 & 0 \\ 0 & x^2 \end{pmatrix}.$$

Then $\text{rank } g_x = 2$ for $x \neq 0$ and $\text{rank } g_0 = 1$. The rank-two strata are $(-\infty, 0)$ and $(0, \infty)$, and the rank-one stratum is $\{0\}$. At the singular point,

$$K_0 = \text{span}\{e_2\}, \quad \bar{E}_0 \cong \mathbb{R}[e_1].$$

Define

$$\rho_x(ae_1 + be_2) = xa \partial_x.$$

For $x \neq 0$ the image lies in the tangent bundle of the corresponding open rank-two stratum. At $x = 0$, $\rho_0 = 0$, so both $K_0 \subseteq \text{Ker } \rho_0$ and $\rho_0(E_0) \subseteq T_0\{0\} = \{0\}$ hold.

In the finite-event hybrid model, choose $x_- < 0$ and $x_+ > 0$. The passage is represented by a reset from x_- to 0 with $J_{2 \rightarrow 1}(a, b) = a$, followed by a reset from 0 to x_+ with $J_{1 \rightarrow 2}(a) = (a, 0)$. This is a hybrid idealization of passage through the rank boundary, not a claim that a compact path lying in one open stratum has 0 as an endpoint. The second component is discarded at entry and is not reconstructed at exit.

14.3 Two noncommuting collapse directions

Let

$$\Pi_x = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad u_\theta = \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}, \quad \Pi_\theta = u_\theta u_\theta^\top.$$

For generic θ ,

$$\Pi_x \Pi_\theta = \begin{pmatrix} \cos^2 \theta & \cos \theta \sin \theta \\ 0 & 0 \end{pmatrix},$$

whereas

$$\Pi_\theta \Pi_x = \begin{pmatrix} \cos^2 \theta & 0 \\ \cos \theta \sin \theta & 0 \end{pmatrix}.$$

Thus $\Pi_x \Pi_\theta \neq \Pi_\theta \Pi_x$, and the final state depends on the order of collapse operations.

14.4 Flat strata with nontrivial hysteresis

Let the base be a closed planar path divided into flat strata. Give every stratum a trivial connection, so every $P_i = \text{Id}$. Insert the collapse and re-embedding maps of the rank-2 $\rightarrow 1 \rightarrow 2$ example. Then every regular curvature vanishes, yet

$$H = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \neq \text{Id}.$$

The memory is generated exclusively by rank transitions.

14.5 External memory restores the discarded component

Enlarge the state by an auxiliary variable ξ and define

$$\tilde{J}_{2 \rightarrow 1}(a, b) = (a, \xi = b).$$

A later activation map may use the stored value:

$$\tilde{J}_{1 \rightarrow 2}(a, \xi) = (a, \xi).$$

The enlarged evolution recovers (a, b) , but recovery does not occur inside the one-dimensional effective fiber alone. This identifies precisely where additional information must enter.

14.6 A non-flat regular DRM

Let

$$M = \mathbb{R}^2, \quad E = TM, \quad \rho = \text{Id},$$

and set

$$\phi(x, y) = x^2 + y^2, \quad g = e^{2\phi}(dx^2 + dy^2).$$

There is one rank-two stratum. The DRM connection is the Levi-Civita connection and the Gaussian curvature is

$$K = -e^{-2\phi} \Delta \phi = -4e^{-2(x^2+y^2)}.$$

This verifies that a regular DRM reproduces ordinary curved Riemannian geometry.

14.7 Energy-amplifying activation

Let $V_- = \mathbb{R}$ and $V_+ = \mathbb{R}^2$ with Euclidean metrics. Define

$$J_\alpha(a) = (a, \alpha a).$$

Then

$$\|J_\alpha a\|_+^2 = (1 + \alpha^2)a^2.$$

For $\alpha \neq 0$, the transition is expansive. The additional energy must be supplied by an external source or represented by an activation cost.

15 Reduction Theorems

Theorem 15.1 (Riemannian reduction). *Suppose:*

- (i) M is an n -manifold;
- (ii) $E \rightarrow M$ has rank n ;
- (iii) g is positive definite everywhere;
- (iv) $\rho : E \rightarrow TM$ is a vector-bundle isomorphism;
- (v) the DRM connection is metric-compatible and torsion-free.

Then DRM induces a Riemannian metric h on M , and the transferred DRM connection is the Levi–Civita connection of h .

Proof. Since g is positive definite, $K = 0$ and $\bar{E} = E$. Define

$$h_p(X, Y) = g_p(\rho_p^{-1}X, \rho_p^{-1}Y).$$

This is a smooth Riemannian metric. Transfer the connection through ρ . The resulting connection on TM is metric-compatible and torsion-free; uniqueness of the Levi–Civita connection gives the result. $\square$

Corollary 15.1 (Sub-Riemannian reduction). *On a constant-rank stratum, if the effective anchor is injective, its image*

$$\mathcal{H} := \text{im } \bar{\rho} \subseteq TS$$

is a constant-rank distribution. Transporting $\bar{g}$ through the anchor defines a sub-Riemannian metric on $\mathcal{H}$.

Corollary 15.2 (Fisher–Rao reduction). *Let $M = \Theta$ be a regular n -dimensional parameter manifold. Assume the effective anchor identifies $\bar{E}_\theta$ with $T_\theta\Theta$ and*

$$\bar{g}_\theta(e_i, e_j) = \mathbb{E}_\theta[\partial_i \log p(X; \theta) \partial_j \log p(X; \theta)].$$

If the connection is metric-compatible and torsion-free, the effective DRM is the Fisher–Rao statistical manifold.

16 Interpretation and Potential Applications

DRM separates three quantities that are often conflated:

1. *ambient capacity*: the bundle rank N ;
2. *effective dimension*: the metric rank $d_{\text{DRM}}(p)$;
3. *observable mobility*: the rank of the effective anchor $\bar{\rho}_p$.

These need not agree. A system may retain a large ambient state capacity while its metric rank collapses. It may retain high metric rank while only a few directions are dynamically observable.

Potential interpretations include:

- temporary loss of a mechanical or controllable degree of freedom;
- passage through a non-identifiable statistical stratum;
- collapse and re-expansion of latent representations;
- switching systems in which the order of constraints matters;
- memory induced by dimensional reduction rather than ordinary curvature.

The mathematical framework does not by itself establish any specific physical, cognitive, or computational application. Such claims require an empirical identification procedure for effective fibers, rank boundaries, transition maps, and memory defects.

17 Limitations and Open Problems

Regularity of rank loci. The regular-stratification assumption excludes pathological rank sets. Whitney stratifications, differential spaces, diffeology, or sheaf-theoretic methods may be required in greater generality.

Canonical transition maps. Transition maps are prescribed data. A theory deriving them canonically from local metric degeneration, dynamics, or a variational principle remains open.

Continuous rank-boundary crossings. The finite-event model uses explicit base resets between points in different strata. A genuinely continuous crossing requires one-sided trace fibers, limiting parallel transports, and hypotheses ensuring that those limits exist and are independent of auxiliary trivializations. That extension is not developed here.

Relation to singular foliation holonomy. A detailed comparison with holonomy groupoids of singular foliations is still needed. DRM explicitly admits non-invertible interstratum maps, but the exact categorical relationship deserves further study.

Nonlinear transitions. The present theory is linear at the effective-fiber level. Nonlinear reset maps require differential, tangent, or categorical extensions.

Infinite transition sequences. Zeno behavior may produce infinitely many rank changes in finite time. Compactness and regularity hypotheses are needed to define limiting transport.

Hybrid geodesics. Existence and uniqueness depend on transition costs, boundary geometry, and reset rules. No general theorem is claimed here.

Experimental identification. Applications require estimators for effective rank, transition maps, energy defects, and order-dependent loop operators from finite noisy data.

Toroidal closure. Boundedness and recurrence do not imply toroidal topology. A torus arises only under additional hypotheses, such as independent commuting complete phase flows with a full-rank period lattice. This is logically separate from the rank-transition holonomy theory.

18 Conclusion

Directional Relational Manifolds provide a stratified geometry for systems whose effective relational dimension changes with state. The foundational data are

$$\mathfrak{D} = (M, E, g, \rho, \{S_\alpha\}, \{\nabla^\alpha\}, \{J_e\}),$$

where E is a fixed-rank ambient bundle, g is a smooth positive-semidefinite metric, ρ is an anchor, S_α are constant-rank strata, ∇^α are stratumwise metric connections, and J_e are rank-transition maps.

The effective fiber and dimension are

$$\overline{E}_p = E_p / \text{Ker } g_p, \quad d_{\text{DRM}}(p) = \text{rank } g_p.$$

Inside a stratum, ordinary metric vector-bundle geometry applies. Between rank strata, explicit transition events replace a nonexistent globally smooth fixed-rank connection.

For a closed hybrid loop,

$$H_\gamma = P_m J_{m-1} P_{m-1} \cdots J_1 P_1$$

acts on the initial effective fiber. If one transition has rank below the initial dimension, then H_γ is rank-deficient, non-invertible, and cannot equal the identity. The system may therefore return to the same base point and the same nominal dimension without recovering its initial relational state. This is geometric hysteresis induced by dimensional change.

The phenomenon can survive when every regular stratum is flat, showing that rank history is a source of path dependence distinct from ordinary curvature. Under dissipativity, the loop operators form a contraction semigroup and admit rank, norm, energy, spectral, and order-sensitive memory invariants.

Under constant-rank and compatibility hypotheses, DRM reduces to established Riemannian, sub-Riemannian, and Fisher–Rao geometries. Its proposed contribution is the integrated treatment of variable metric rank, quotient geometry, anchored motion, hybrid transition maps, and non-invertible global transport.

A Local Matrix Criteria

In a local frame $e_1, \dots, e_N$, let

$$G(p) = (g_p(e_i, e_j)).$$

Then

$$d_{\text{DRM}}(p) = \text{rank } G(p), \quad K_p = \{z^i e_i : G(p)z = 0\}.$$

On a rank- r stratum, an adapted frame gives

$$G(p) = \begin{pmatrix} G_{\text{eff}}(p) & 0 \\ 0 & 0 \end{pmatrix},$$

where $G_{\text{eff}}(p)$ is positive definite.

For a transition represented by J , dissipativity and isometry are respectively equivalent to

$$J^\top G_+ J \preceq G_-, \quad J^\top G_+ J = G_-.$$

For a closed loop with initial metric G , the energy-defect matrix is

$$D_\gamma = G - H_\gamma^\top G H_\gamma.$$

The loop is dissipative exactly when $D_\gamma \succeq 0$.

B Numerical Diagnostics

For finite precision, one may estimate effective rank by

$$d_\varepsilon(G) = \#\{\lambda_i(G) > \varepsilon\}.$$

A spectral effective rank is

$$r_{\text{eff}}(G) = \exp\left(-\sum_i p_i \log p_i\right), \quad p_i = \frac{\lambda_i}{\sum_j \lambda_j},$$

for nonnegative eigenvalues with nonzero sum.

When G_- is positive definite on the effective incoming fiber, define

$$\delta_J = \lambda_{\max}\left(G_-^{-1/2} J^\top G_+ J G_-^{-1/2}\right) - 1.$$

Then $\delta_J \leq 0$ characterizes dissipativity.

Useful loop indices include

$$h_{\text{rank}} = 1 - \frac{\text{rank } H_\gamma}{n}, \quad h_F = \frac{\|H_\gamma - \text{Id}\|_F}{\sqrt{n}},$$

and, for $v \neq 0$,

$$h_{\text{energy}}(v) = 1 - \frac{\|H_\gamma v\|^2}{\|v\|^2}.$$

C Conditional Toroidal Closure

The rank-transition framework does not imply toroidal topology. For completeness, a sufficient condition is recorded.

Theorem C.1 (Conditional toroidal closure). *Let A be a compact connected invariant manifold admitting n complete, pointwise linearly independent, commuting vector fields $X_1, \dots, X_n$. Suppose their joint $\mathbb{R}^n$ -action is transitive and the stabilizer of a point is a full-rank lattice $\Lambda \subset \mathbb{R}^n$. Then*

$$A \cong \mathbb{R}^n / \Lambda \cong \mathbb{T}^n.$$

Proof. The commuting complete fields generate a transitive action of $\mathbb{R}^n$ on A . Thus A is diffeomorphic to the homogeneous quotient $\mathbb{R}^n / \text{Stab}(p) = \mathbb{R}^n / \Lambda$. A quotient of $\mathbb{R}^n$ by a full-rank lattice is diffeomorphic to the n -torus. $\square$

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Conflict of Interest

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